



**Sacramento River Bank Protection Project
Long-Term Aquatic Monitoring
Fish Sampling and Habitat Characterization
Fiscal Years 2011 through 2015**

Supplement to the 2014 Annual Report

Prepared for
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This supplement is intended to accompany the 2014 Annual Report (FISHBIO 2015), which contains, in great detail, the pertinent background, applicable field collection, and data analysis methods, as well as occupancy model selections.

Executive summary

- Four reaches in the vicinity of Verona (RM 78) and the Fremont Weir (RM 83; each 1,000 meters long) were sampled by electrofishing in late March and early April, 2015.
- Reaches encompassed a range of habitat features and conditions, capturing habitat diversity not represented at single locations during past monitoring.
- The number of observed juvenile salmonids was very low (5% of points sampled), likely due to the timing of field sampling (later than in previous years).
- Point-specific presence/absence records of juvenile Chinook salmon were used to evaluate the performance of a previously developed occupancy model, including the variables bankslope, velocity gradient, instream woody material, velocity & substrate (both 10 feet from the wetted margin).
- The overall model performance was adequate, with over 85% of occupancy records classified correctly (Cohen's kappa; $p = 0.04$).
- Over 95% of absences were predicted correctly, and about 35% of observed presence records were predicted correctly.
- The relatively high error rate is likely attributable to the different densities of Chinook salmon in the study reach during sampling in 2015, as compared to earlier years (data upon which the model was built, with about 20% occupancy).
- Despite drastically differing abundance and differences in environmental conditions, the key factors of bankslope, velocity gradient, instream woody material, velocity 10 feet from the wetted margin, and substrate 10 feet from the wetted margin should be considered important habitat characteristics determining habitat use by juvenile Chinook salmon in the study area.
- For smallmouth bass, the most abundant predatory species observed during 2015 sampling, the final predictive model formulated based on a multiyear dataset (FISHBIO 2015) had to be modified, as it included site category as a predictive variable (which was not available for the 2015 data collection)
- Following exclusion of this variable, correct classification of occupancy records was 25%, only a slight improvement over random classification.

Background and introduction

As part of an ongoing assessment of post-construction performance of various levee repair designs on the Sacramento River, FISHBIO has conducted fish- and habitat surveys to evaluate habitat use of focal fish species at post-2006 levee repair sites over the past four years. These sites contain designed mitigation features, which are intended to improve fish habitat over historically predominant, non-mitigated rock revetment repairs. With guidance, oversight, and collaboration by an interagency work group, with members representing the USACE, DWR, USFWS, CDFW and NMFS, monitoring methodologies have been refined over the course of the past several years to better address pertinent monitoring and research questions. Fisheries monitoring, conducted primarily during the rearing and peak outmigration season of anadromous salmonids (January through April), has resulted in valuable insights regarding fish habitat use at differing repair designs over the course of the past years (see FISHBIO 2012 – 2015). In general, this work has shown that abundance of juvenile salmonids is higher at mitigated repair sites compared to non-mitigated (rock revetment) reference locations, and, for some mitigation designs, approximates or exceeds habitat use at naturalized sites (FISHBIO 2015).

Furthermore, consistent collection of detailed habitat characteristics at sampling points has permitted identification of microhabitat features that significantly affect habitat occupancy by focal fish species (and life stages). Identification of these habitat characteristics has permitted the formulation of relatively simple habitat occupancy models that have performed well in correctly predicting presence/absence of target species at specific points along monitored sites (generally upwards of 85% correct classification using an unbiased jackknife procedure; FISHBIO 2015). The models perform particularly well for Chinook salmon fry and juveniles, and for smallmouth bass, the most abundant introduced predatory species in the study reach. In addition, monitoring has shown that juvenile Chinook salmon actively rear in the project reach, as evidenced by (sometimes extensive) feeding.

While these results suggest that near-shore riverine habitat within the footprint of bank protection features can be enhanced through implementation of alternative repair design and incorporation of mitigation features, targeted evaluation of specific features (such as deliberately placed large woody material) has been hampered by low river flows, a consequence of ongoing, severe drought conditions. As such, the 2015 field sampling effort was intended to:

- Evaluate the performance of habitat occupancy models established over the course of the monitoring effort by validating their applicability to shoreline habitat in the project reach, using an independent dataset that has not been included in model formulation; and,
- Provide complimentary insights regarding habitat use by juvenile salmonids, as pertaining to specific habitat features such woody material or inundated vegetation that have been poorly represented in the existing dataset due to lower-than-average runoff conditions and associated lack of inundation/activation of mitigation features.

Spring 2015 sampling

Two sampling events were conducted during the spring of 2015. During each event, two 1,000-meter reaches were sampled along both banks (river left [RL] and river right [RR]). These reaches were selected by the USACE and FISHBIO to be representative of a broad range of shoreline categories, and included sections with bank stabilization features and naturalized sections.

The downstream sampling reach was located near RM 78, hereafter referred to as Verona (just downstream of the Feather River confluence; N38.75942°, W121.59203° to N38.76962°, W121.59550°) and was sampled on 3/23/2015 and 4/8/2015. The upstream sampling reach was located by the Fremont Weir near RM 83 (N38.76365°, W121.66501° to N38.76049°, W121.67369°), and was sampled on 3/24/2015 and 4/2/2015 (see Figure 1 for corresponding river discharge/stage).

The sampling methodology (i.e., fish capture, fish handling, assessment of habitat characteristics, etc.) has remained consistent with the methodology described in detail in annual project reports for 2013 and 2014 (FISHBIO 2014 and FISHBIO 2015).

With respect to habitat monitoring efforts, we used an ARIS camera, paired with a regular video camera (GoPro), to provide a continuous visual representation of nearshore habitat in the sampled reaches above and below the waterline. Resulting dual footage was submitted to the USACE in May 2015.

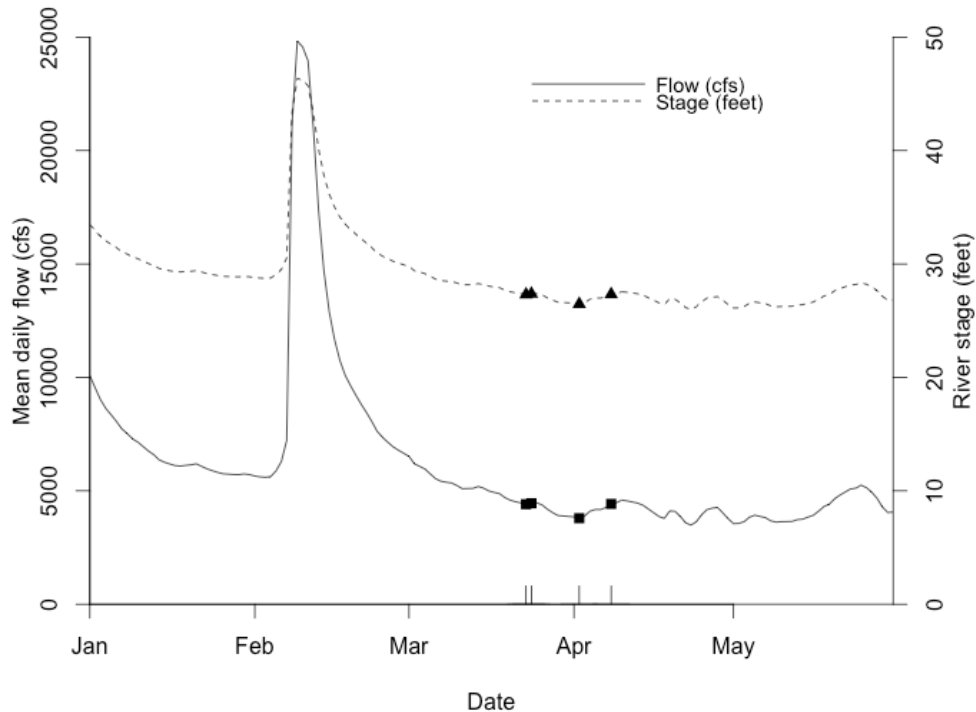


Figure 1. Flow (cfs) and river stage (feet) on the Sacramento River at Wilkins Slough (WLK) during spring 2015.

Results - Species captured

A total of 17 fish species were collected at the above-specified sampling locations in 2015 (Table 1). Overall, focal fish species recorded (captured and missed) included 33 juvenile Chinook salmon (of 33 captured individuals, one was classified as winter-run, 19 as spring-run and the remaining 13 as fall-run; also see Table 2) and one steelhead (of hatchery origin). Notably, all captured Chinook salmon were larger than 55 mm FL (i.e. no Chinook salmon were classified as “fry”; Figure 2). No mortalities of Chinook salmon or steelhead occurred.

Predatory species included 74 smallmouth bass, one striped bass, and 53 Sacramento pikeminnow. In addition, 56 inland silversides (*Menidia beryllina*)¹ were observed during the 2015 sampling effort. Take of predatory species (including inland silverside) is summarized in Table 3. Length frequency distributions by sampling event for smallmouth bass and Sacramento pikeminnow are depicted in Figure 3.

Table 1. Summary of fish species captured in 2015².

Common (Scientific) Names	Native Status (N/ I) ¹	Count	Mean Length ² (mm)
Focal Species			
Chinook salmon (<i>Oncorhynchus tshawytscha</i>)	N	33	77
Steelhead/rainbow (<i>Oncorhynchus mykiss</i>)	N	1	263
Predatory Species			
Sacramento pikeminnow (<i>Ptychocheilus grandis</i>)	N	53	106
Striped bass (<i>Morone saxatilis</i>)	I	1	475
Smallmouth bass (<i>Micropterus punctulatus</i>)	I	74	140
Inland silverside (<i>Menidia beryllina</i>)	I	56	78
Other species			
Threadfin shad (<i>Dorosoma petenense</i>)	I	55	98
Hardhead (<i>Mylopharodon conocephalus</i>)	N	10	181
Red shiner (<i>Cyprinella lutrensis</i>)	I	46	37
Sacramento sucker (<i>Catostomus occidentalis</i>)	N	41	57
Channel catfish (<i>Ictalurus punctatus</i>)	I	1	350
White catfish (<i>Ameiurus catus</i>)	I	2	62
Black crappie (<i>Pomoxis nigromaculatus</i>)	I	1	101
Warmouth (<i>Lepomis gulosus</i>)	I	4	89
Green sunfish (<i>Lepomis cyanellus</i>)	I	12	87
Bluegill sunfish (<i>Lepomis macrochirus</i>)	I	4	80
Redear sunfish (<i>Lepomis microlophus</i>)	I	11	114

¹ N = native, I = introduced; ² Lengths are provided as fork lengths (mm) for all species with forked or truncated caudal fins, otherwise total length (mm) was used.

¹ Inland silversides are potential predators of delta smelt eggs and/or larvae. However, sampling was not conducted in delta smelt spawning habitat downstream of RM 20, so these fish are unlikely to function as predators in the sampling reach. Nonetheless, data was provided regarding their presence in sampled sites.

² See earlier reports (e.g. FISHBIO 2014, FISHBIO 2015) for a summary of species captured during earlier years.

Table 2. Summary of juvenile Chinook salmon captures during 2015 monitoring. No mortalities occurred (denoted in parentheses).

Sampling Event:	1 (3/23 & 3/24)				2 (4/2 & 4/8)				Total (no. morts.)	Mort %
Location:	Fremont		Verona		Fremont		Verona			
Bank (L/R):	L	R	L	R	L	R	L	R		
Fall	1	2	0	2	5	0	0	3	13 (0)	0%
Late-fall	0	0	0	0	0	0	0	0	0	0%
Spring	3	0	3	1	4	2	1	5	19 (0)	0%
Winter	1	0	0	0	0	0	0	0	1 (0)	0%
Total	5	2	3	3	9	2	1	8	33 (0)	0%

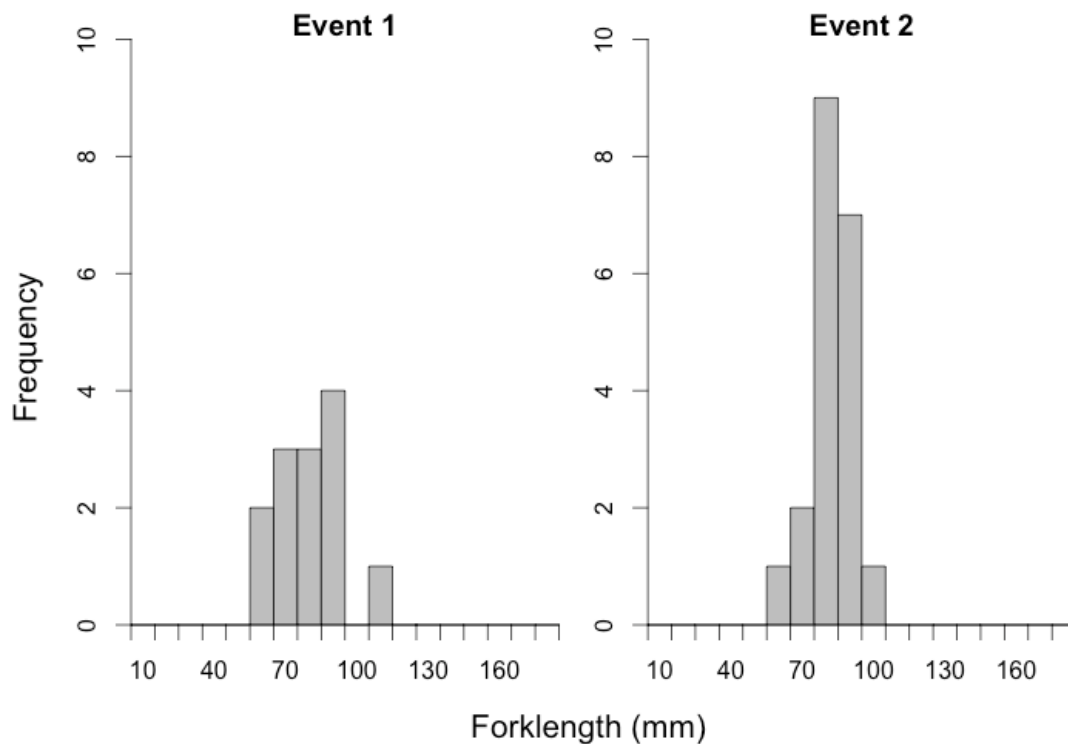


Figure 2. Length-frequency histograms of Chinook salmon captured during 2015 monitoring (four sampling locations combined for each event; see text for sampling dates).

Table 3. Summary of sampled predatory species, by sampling event, during spring 2015 monitoring. Numbers of individuals that exceeded 150mm FL are given in parentheses.

Sampling Event	Smallmouth bass	Striped bass	Sacramento pikeminnow	Inland silverside	Total
1	34 (11)	1 (1)	32 (4)	15	82
2	43 (12)	0	23 (1)	41	107
Total	77 (23)	1 (1)	55 (5)	56	189

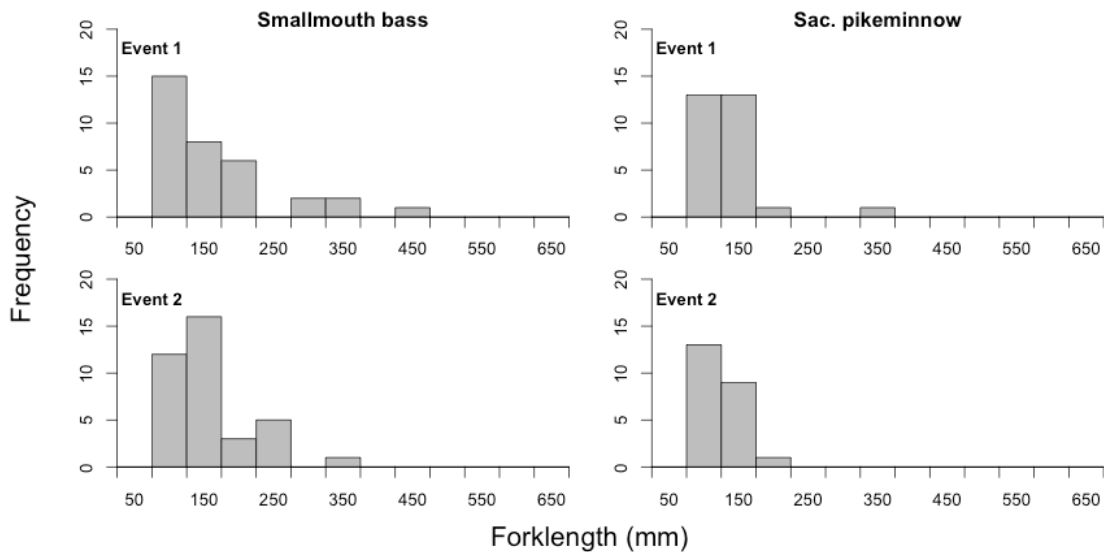


Figure 3. A comparison of length-frequency histograms of predominant piscivorous species for the two sampling events conducted in 2015 (by row).

Juvenile salmon presence/absence and applicability of the occupancy model

A total of 556 point-specific presence/absence samples were collected in 2015, representing samples for four sampling sites (each sampled twice). Chinook salmon fry were not observed in 2015, likely due to sampling later than in previous years (when most fry-sized Chinook were collected in February and early March). Chinook salmon juveniles were observed in less than 6% of the samples (31/556).

Rather than formulating a logistic regression model that describes the occupancy probability of juvenile Chinook salmon based on data collected in 2015, this year's presence/absence records were used to evaluate the applicability of the habitat occupancy model based on cumulative monitoring data (i.e., a predictive model based on over 2,000 occupancy records collected during 2013 and 2014 surveys; see Section 6.1.4. in FISHBIO 2015).

Predicted probabilities of occupancy were estimated using habitat data collected during 2015 surveys. The coefficients and variables included in the final predictive model for occupancy based on 2013-2014 are summarized in Table 4. To assess the classification accuracy of the occupancy, model predictions of occupancy were compared against field observations of occupancy. Overall, more than 80% (86.15%) of occupancy records were classified correctly, using the five predictive variables listed in Table 4. The significance value of Cohen's kappa statistic was 0.044, indicating that the model provided a significant improvement over random classification. The majority of correct classification is attributable to correctly predicted absence (of the 457 locations where absence was predicted, juvenile Chinook salmon were absent at 437, corresponding to a false positive error rate of 4.38%). Of the 31 observed presences, 11 (35.48%) were predicted correctly, which constitutes a significant improvement over random chance, as indicated by the above mentioned kappa statistic.

Table 4. Coefficients for final multivariate logistic regression model fit predicting the probability of Chinook salmon juvenile occupancy, based on multiple years of occupancy and habitat monitoring (see FISHBIO 2015 for details). Definitions of explanatory variables: instreamWM = instream woody material; vel10 = velocity 10 feet from the wetted margin; and substrate10 = substrate classification 10 feet from the wetted margin – default = Mud, R = Rock, and G = Gravel.

Variable	Coefficient estimate	Std. error	<i>p</i>	Odds Ratio
Intercept	-3.82	0.80	<0.01	
bank slope	-0.88	0.55	0.10	0.41
Velocity gradient	-10.83	3.82	<0.01	<0.01
instreamWM				
Sparse	0.79	0.14	0.08	2.23
Medium	0.80	0.23	<0.01	2.31
Dense	-0.25	0.33	0.45	0.81
Vel10				
V2	0.03	0.21	0.88	1.01
V3	0.67	0.38	0.08	1.92
Substrate10				
R	-0.23	0.14	0.11	0.80
G	-0.46	0.52	0.37	0.63

It should be considered that the dataset upon which the model was built contained approximately 20% presence records, as compared to less than six percent of occupied points during 2015 data collection. As a consequence, discrepancies between model predictions and observed data are expected. Though this model was better at predicting absence (95.62% correct) than presence (35.48% correct), it can be concluded that - despite differing abundance and differences in environmental conditions - the variables identified in the 2014 annual report (bankslope, velocity gradient, instream woody material, velocity 10 feet from the wetted margin, and substrate 10 feet from the wetted margin; FISHBIO 2015) should be considered important habitat characteristics determining the presence or absence of juvenile Chinook salmon in the study area. As a consequence, such features in the pertinent range of values should be considered for inclusion in future bank repair planning if localized hydraulic constraints permit.

Smallmouth bass presence/absence and applicability of the occupancy model

As indicated previously, a total of 556 point-specific presence/absence samples were collected in 2015, representing samples for four sampling sites (each sampled twice). Smallmouth bass (all sizes combined) were observed at 11.3% of sampling points (63/556).

Contrary to the predictive model formulated using multi-year occupancy data for juvenile Chinook salmon (FISHBIO 2015), the previously derived smallmouth bass occupancy model included site category (referring to repair design), which was considered the most important predictive variable. As data collected in 2015 was not specific to any given

repair design, but rather encompassed long reaches of shoreline representative of a range of environmental conditions, the previously defined model was modified to exclude site category as a predictor.

As a consequence, the most influential predictive variable was absent from this classification model, which is summarized in Table 5.

Table 5. Coefficients for final multivariate logistic regression model fit predicting the probability of smallmouth bass occupancy, based on multiple years of occupancy and habitat monitoring (see FISHBIO 2015 for details). Definitions of explanatory variables: instreamVeg = instream vegetation; instreamWM = instream woody material; DepthX = depth x feet from the wetted margin; velX = velocity x feet from the wetted margin; and substrateX = substrate classification x feet from the wetted margin – default = Mud, R = Rock, and G = Gravel.

Variable	Coefficient estimate	Std. error	<i>p</i>	Odds Ratio
Intercept	0.17	0.09	0.06	
RM	-0.003	0.0004	<0.01	0.99
Shade				
present	0.24	0.02	<0.01	1.27
Velocity gradient	-1.41	0.39	<0.01	0.24
Temp. difference	0.03	0.02	0.04	1.03
Depth5				
Shallow2	0.04	0.02	0.05	1.04
Shallow3	0.09	0.03	<0.01	1.09
Deep	0.09	0.05	0.08	1.09
Depth15				
Shallow2	-0.06	0.09	0.50	0.94
Shallow3	0.01	0.09	0.93	1.01
Deep	0.03	0.09	0.78	1.03
Vel15				
Medium	0.07	0.02	<0.01	1.08
Fast	0.10	0.03	<0.01	1.11
instreamVeg				
Sparse	0.001	0.02	0.95	1.001
Medium	0.03	0.03	0.19	1.03
Dense	-0.05	0.03	0.06	0.95
instreamWM				
Sparse	0.05	0.01	<0.01	1.05
Medium	0.02	0.02	0.51	1.02
Dense	0.03	0.03	0.34	1.03
Substrate5				
R	0.04	0.02	0.02	1.04
G	-0.03	0.04	0.45	0.97
Substrate10				
R	0.06	0.02	<0.01	1.07
G	0.06	0.05	0.19	1.06

Overall, about 25% of occupancy records were classified correctly, using the predictive variables listed in Table 5. The significance value of Cohen's kappa statistic was 0.12, indicating that the model provided a slight improvement over random classification. Of 448 predicted presences, 60 (13%) were classified correctly. Conversely, of the 63 points that were occupied by smallmouth bass, 60 (95%) were predicted correctly.

Clearly, while the model formulated herein (constituting a change from the "best" model identified in FISHBIO 2015) resulted in a slight improvement over "random" classification, the overall model performance is rather poor. As indicated previously, the inclusion of site category in the best model fit identified in FISHBIO 2015 was a major contributor to the nearly 88% correct classification (jackknife validation) for the multi-year dataset used to formulate the predictive model.

Consequently, while

Literature cited

- FISHBIO. 2012. Sacramento River Bank Protection Project Long Term Aquatic Monitoring: Fish Sampling and Habitat Characterization Annual Report 2011. Prepared for United States Army Corps of Engineers, Sacramento (CA).
- FISHBIO. 2013. Sacramento River Bank Protection Project Long Term Aquatic Monitoring: Fish Sampling and Habitat Characterization Annual Report 2012. Prepared for United States Army Corps of Engineers, Sacramento (CA).
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